Code No: 157FB **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
UTILIZATION OF ELECTRICAL ENERGY
(Common to CSE, IT, ITE)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a) What are the causes of failure of heating element? [2]
b) What is dielectric heating? Explain its applications. [3]
c) In the arc welding, What is the order of temperature of the arc produced? [2]
d) How Seam welding will be implemented in industrial applications than Spot welding. Compare the both welding with suitable description. [3]
e) A surface inclined at an angle 40^0 to the rays is kept 6 m away from 150 candle power lamp. Find the average intensity of illumination on the surface. [2]
f) Define Solid angle and Space height ratio with respect to illumination. [3]
g) Give the advantages and disadvantages of electric traction over other types of traction systems. [2]
h) Assume Kakatiya past passenger train is running at an average speed of 42 kmph between stations situated 2.6 km apart. Train accelerates at 3.4 kmphps and retards at 2 kmphps. Find its maximum speed assuming simplified trapezoidal speed time curve. [3]
i) State the need for double battery parallel block system used in electric fraction. [2]
j) Elaborate the coaching wiring used in electric fraction. [3]

PART – B

(50 Marks)

2.a) Describe briefly the methods of direct and indirect resistance heating and list the applications of these two methods?
b) A20-kW, 230-V, and single-phase resistance oven employs nickel-chrome strip 25-mm thick is used, for its heating elements. If the wire temperature is not to exceed $1,200^0$C and the temperature of the charge is to be 700^0C. Calculate the width and length of the wire. Assume the radiating efficiency as 0.6 and emissivity as 0.9. Determine also the temperature of the wire when the charge is cold [5+5]

OR

3.a) How will you control most efficiently the heat of resistance furnace? What advantages does graphite electrode process over carbon electrode?
b) State the advantages of electrically produced heat by means of arc furnaces. Distinguish between the direct and indirect type of arc furnaces with applications [5+5]

4.a) Discuss the comparison between Resistance and Arc Welding with salient points.
b) How would you describe the Electrolysis process with neat circuit diagram in Electric welding? [5+5]

OR

5.a) Inference between Carbon Arc welding and Metal Arc welding methods with neat diagrams. Also give their applications.
b) How would you describe Electric arc welding with neat sketch? [5+5]

6.a) The luminous intensity of a lamp is 600 CP. Find the flux given out. Also find the flux in the hemisphere containing the source of light and zero above the horizontal.

b) Analyze the factors which required for proper illumination to the following:
i) Street lighting
ii) Factory lighting
iii) Flood lighting. [5+5]

OR

7.a) State the laws of illumination. Explain the laws with the help of suitable diagrams, and derive an equation of the same.

b) An illumination on the plane of 220 lux is required in a room 86 m×20 m in size. The lamps are required to be hung 3.8 m above the surface. Assuming a suitable space-height ratio, a utilization factor of 0.76, a lamp efficiency of 42 lumens per watt and a depreciation factor of 1.55, Estimate the number of lamps, rating of each lamp and also dispose the lamps in the room. [5+5]

8.a) Derive expression for the distance traveled by an electric train using trapezoidal speed-time curve?

b) An electric train weighing 250 tonnes runs a uniform up gradient of 1% with the following speed-time curve:
i) Uniform acceleration of 2.3 kmphps for 45 sec
ii) Constant speed of 2.1 kmphps for 34 sec
iii) Coasting for 32 sec
iv) Braking at 2.9 kmphps to rest
v) Stop at station 22 sec
If the tractive resistant is 44 N/tonne, rotational inertia effect 10% of dead weight and overall efficiency of transmission and motor is 78%.
Determine
i) Schedule speed
ii) Specific energy consumption
iii) Total energy consumption
iv) Distance between two stations. [5+5]

OR

9.a) What are the different methods of approximation of speed-Time curves? Derive expression for distance travelled using quadrilateral approximation method of V (t) curve.

b) A train has a schedule speed of 30kmph over a level track. Distance between stations being 1km. Station stopping time is 20 seconds. Assuming braking retardation of 3 kmphps and maximum speed 25 percentage greater than average speed, calculate acceleration required to run the service. [5+5]

10. With the help of neat diagram, explain the method to obtain unidirectional polarity in train lighting. [10]

OR

11. Explain the special requirements of a train lighting. [5+5]

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